

Roll No.

TBT-603(C)

B. TECH. (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

RATIONAL DRUG DISCOVERY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the flowchart for drug discovery process in detail.

10 Marks (CO1)

OR

- (b) Discuss structure based drug designing. What are the tools used in SBDD ?

10 Marks (CO1)

2. (a) What is Lipinski's rule of 5 ? How is it related to ADMET parameters ?

10 Marks (CO1)

OR

- (b) Explain various strategies for drug target identification and validation.

10 Marks (CO1)

P. T. O.

(2)

3. (a) Differentiate between Molecular mechanics and Quantum mechanics.

10 Marks (CO2)

OR.

- (b) What is Molecular Modeling ? Explain the applications of combinatorial chemistry.

10 Marks (CO2)

4. (a) Describe force fields. Explain the types of parameters considered in force field.

10 Marks (CO2/CO3)

OR

- (b) Explain Steepest Descent and Newton Raphson algorithms for Energy minimization.

10 Marks (CO2/CO3)

5. (a) Discuss various types of Ensembles used in GROMACS during molecular modeling simulation.

10 Marks (CO3)

OR

- (b) Describe steps of principle component analysis. What are applications of PCA ?

10 Marks (CO3)

Roll No.

TBT-603(B)

B. TECH. (BIOTECHNOLOGY) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

PYTHON

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What is Python ? Discuss the features and importance of Python.

10 Marks (CO1, CO2)

OR

- (b) What is a data type ? Describe the basic data types in python.

10 Marks (CO1, CO3)

2. (a) What is an expression ? Explain the precedence of operators in Python with suitable code example.

10 Marks (CO1, CO4)

OR

- (b) What are a String ? Explain with various functions/methods used with Strings in Python.

10 Marks (CO1, CO4)

3. (a) What is an indexing and slicing in python ? Write a python code to print a given string in reverse using the indexing.

10 Marks (CO2, CO5)

P. T. O.

(2)

OR

(b) Write a prime number ? Write a Python program to check if a given number is prime or not. 10 Marks (CO3, CO4)

4. (a) What are the python loops ? Illustrate the use of break and continue keyword in python. 10 Marks (CO3, CO5)

OR

(b) Write a python program to print the first 10 numbers of Fibonacci sequence. 10 Marks (CO2, CO5)

5. (a) What is Python casting ? Describe the constructor functions used for casting in Python. 10 Marks (CO1, CO5)

OR

(b) Write a Python program to input N numbers. Then write a code to print sum, average (arithmetic mean), maximum, and a minimum of the input values. 10 Marks (CO2, CO4)

Roll No.

TBT-602

B. TECH. (BIOTECHNOLOGY) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

PLANT BIOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What is Micropropagation ? Discuss its advantages and disadvantages.

10 Marks (CO1)

OR

- (b) What are somaclonal variations ? How would you use it for your benefit in tissue culture experiments ?

10 Marks (CO1)

2. (a) Explain in detail the principle and methodology for axillary bud proliferation.

10 Marks (CO1)

OR

- (b) With the help of a diagram, discuss the various stages in the micropropagation of plants.

10 Marks (CO1)

3. (a) Explain in detail how we can produce in vitro plantlets using indirect somatic organogenesis.

10 Marks (CO1, CO2)

P. T. O.

(2)

OR

(b) Explain the following terms in detail : 10 Marks (CO1, CO2)

(i) Synthetic seeds

(ii) Cryotherapy

4. (a) What is androgenesis ? Explain in detail the methodology for anther culture. 10 Marks (CO2)

OR

(b) With the help of a diagram, explain the protocol for in vitro microspore culture. 10 Marks (CO2)

5. (a) Explain the following terms in detail : 10 Marks (CO2)

(i) Gynogenesis

(ii) Protoplasts

OR

(b) What is cybridization ? Explain with the help of a diagram.

10 Marks (CO2)

Roll No.

TBT-601

B. TECH. (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

ANIMAL BIOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Brief the future scope of Animal biotechnology. 10 Marks (CO1)

OR

- (b) What is role of serum in animal cell culture ? 10 Marks (CO1)

2. (a) What is cryopreservation ? Describe its role and technique. 10 Marks (CO1)

OR

- (b) What difference between the culture and transport media ? 10 Marks (CO1)

3. (a) What do you think about livestock biotechnology ? Brief. 10 Marks (CO2)

OR

- (b) What is transgenic animals ? Give some examples. 10 Marks (CO2)

P. T. O.

(2)

4. (a) Describe the methods of transgenesis. 10 Marks (CO2)

OR

- (b) What is IVF ? Describe its role in livestock production. 10 Marks (CO2)

5. (a) Describe the superovulation in detail. 10 Marks (CO2)

OR

- (b) What is the role of embryo transfer technique in livestock production ?

10 Marks (CO2)

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TBT-604(B)

B. TECH. (BIOTECHNOLOGY)

(SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

NANOBIOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Discuss in brief about NANOFIBRES and its applications.

10 Marks (CO1)

OR

(b) Explain in brief about graphene based materials and its applications.

10 Marks (CO1)

2. (a) Differentiate between top-down and bottom-up approaches.

10 Marks (CO2)

OR

(b) Summarize in brief biological production of metal nanoparticles.

10 Marks (CO2)

P. T. O.

(2)

3. (a) Discuss in brief about nanoparticles and its role in biomedical research.

10 Marks (CO3)

OR

- (b) Summarize in brief about the mechanism of nano drug delivery.

10 Marks (CO3)

4. (a) Explain in brief about microfabrication techniques. 10 Marks (CO1)

OR

- (b) Describe in brief about biological motors or biological machines.

10 Marks (CO2)

5. (a) Discuss in brief about nanoplates and its application. 10 Marks (CO1)

OR

- (b) Describe its history of nanobiotechnology. 10 Marks (CO1)